Black box Test Suite Assignment

Function Test Plan and Findings

Version 2.0

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1. Introduction

1.1 Purpose

Develop a set of black-box test cases for a clothing retailer that asks user for information and then calculates and displays the cost of the article of clothing. The tests should be Act-Like-A-Customer tests and the goal is to have enough test cases to cover all possible (discrete) inputs for the inputs that are not numbers (continuous).

1.2 Scope

The scope of this assignment encompasses the execution, design as well as evaluation of black-box test cases for a web-based pricing application. The objective is to ensure that the application functions properly in its calculations of prices based on user inputs while also handling different boundaries, conditions and combinations of inputs.

2. System Overview

The system is an online web application for a clothing retailer that takes information as user input and calculates and displays the cost as output. The charges of the material that is required to make the shirt are determined by the selected features and dimensions of the shirt, instead of having a flat fee like many other retailers.

3. Requirements

3.1 Software Requirements

The basic functional requirement for such an application is a working User Interface that displays input fields, performs error handling and displays the output. It also uses pricing calculation logic to calculate prices.

Nonfunctional requirements include security, scalability and performance.
Platform requirements include a front-end(client-side) and a back-end(server side).

3.2 Hardware Requirements

The hardware requirements will depend on the complexity, scale and the expected load of the application.

4. Testing

4.1 Test Strategy

Primarily, functional and boundary tests are used for these test cases that cover all possible combinations of long sleeves and collar (Yes/No), both currency options (USD and EUR) as well as extreme and atypical values of input sizes to test how the system handles various scenarios. Edge cases are also used such as zero input and very small and large sizes.

4.2 Proposed Test Cases

4.2.1 Test Case 01: *Basic, No Extras, USD*

Description: A customer selects no long sleeves, no collar and USD currency.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps

1. Open application and go to the shirt customization page.
2. Enter 28 in the waist size field
3. Enter 22 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is unchecked.
5. Ensure that the Collar checkbox is unchecked.
6. Select USD from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

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Expected output: $55.44 USD

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.2 Test Case 02: With Collar, USD

Description: A customer selects collar, currency USD

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 27 in the waist size field
3. Enter 23 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is unchecked.
5. Ensure that the Collar checkbox is checked.
6. Select USD from the currency dropdown.
7. Click Calculate button.

8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: $57.89 USD

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.3 Test Case 03: Long Sleeves, No collar, USD

Description: A customer selects long sleeves, no collar and USD currency.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 26 in the waist size field
3. Enter 24 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is checked.
5. Ensure that the Collar checkbox is unchecked.
6. Select USD from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: $61.16

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.4 Test Case 04: Long Sleeves, Collar, USD

Description: A customer selects long sleeves with collar and USD currency.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 25 in the waist size field
3. Enter 22 in the length from neck to waist field.

4. Ensure that the long sleeves checkbox is checked.
5. Ensure that the Collar checkbox is checked.
6. Select USD from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: $56.50

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.5 Test Case 05: Long Sleeves, Collar, EUR

Description: A customer selects long sleeves with collar and EUR currency.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 22 in the waist size field
3. Enter 24 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is checked.
5. Ensure that the Collar checkbox is checked.
6. Select EUR from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: € 47.97

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.6 Test Case 06: No Long Sleeves, No Collar, EUR

Description: A customer selects no long sleeves, no collar and EUR currency.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 25 in the waist size field
3. Enter 22 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is unchecked.
5. Ensure that the Collar checkbox is unchecked.
6. Select EUR from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: € 43.56

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.7 Test Case 07: Smallest size, USD

Description: A customer selects the smallest possible waist length and size and USD currency.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 1 in the waist size field
3. Enter 1 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is unchecked.
5. Ensure that the Collar checkbox is unchecked.
6. Select USD from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: $0.09

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.8 Test Case 08: Maximum reasonable size, USD

Description: To test the system's handling of larger inputs, a customer enters a large waist size and length and selects USD currency. Also checks long sleeves and collar.

Requirements: The Web Application should be running and the pricing application should

be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 81 in the waist size field
3. Enter 40 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is checked.
5. Ensure that the Collar checkbox is checked.
6. Select USD from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: $298.6

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.9 Test Case 09: Zero waist or length, USD

Description: A customer enters 0 for either the length or waist size (invalid input) and selects USD.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 0 in the waist size field
3. Enter 35 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is unchecked.
5. Ensure that the Collar checkbox is unchecked.
6. Select USD from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: $0

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.10 Test Case 10: Zero waist and length, EUR

Description: A customer enters 0 for both the length and waist size (invalid input) and selects EUR. Checks long sleeves and collar.

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 0 in the waist size field
3. Enter 0 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is checked.
5. Ensure that the Collar checkbox is checked.
6. Select EUR from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

Expected output: €6.16

Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

4.2.11 Test Case 11: Max size, EUR

Description: A customer enters the largest reasonable waist length and size, selects long sleeves and collar and selects EUR currency

Requirements: The Web Application should be running and the pricing application should be fully accessible, developed and deployed. Required functionality should also be enabled such as input fields for length, waist size, checkboxes for long sleeves and collar and an available and functional dropdown for currency.

Prerequisites: User should be logged in to the system.

Steps: User Activity steps:

1. Open application and go to the shirt customization page.
2. Enter 80 in the waist size field
3. Enter 40 in the length from neck to waist field.
4. Ensure that the long sleeves checkbox is checked.
5. Ensure that the Collar checkbox is checked.
6. Select EUR from the currency dropdown.
7. Click Calculate button.
8. Verify that the output is as expected.
9. The test has passed if the output is as expected. Report the discrepancy if this is not the case and there is an error.

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Expected output: €259.6
Assumptions: There is no interference with the test result with the test result due to browser caching or previous session data. The test is executed in a clean environment, the web application handles user input validation correctly and the checkboxes and dropdowns function correctly.

5. Assumptions and Constraints

5.1 Assumptions

The assumptions that guide the functionality and design of this application are that input values are reasonable, currency will be converted correctly, check boxes will be used correctly and there will be no need for user authentication as this is a guest-based system.

5.2 Constraints

Functional constraints include fixed pricing logic, limited customization, currency conversion limitation and no multi-currency support beyond USD/EUR. Technical constraints include performance, scalability, limited error handling and a static front-end. Resource constraints include time and budget constraints, device and browser compatibility and geographical limitations it is presumed that the users will be mainly from regions where either USD or EUR are the accepted currency.